A Probabilistic Model of Motor Subtype Progression in Parkinson's Disease

Project Report

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Abstract

Parkinson's disease is classified into Tremor Dominant (TD), Indeterminate and Postural Instability / Gait Difficulty (PIGD) motor subtypes by a fixed ratio of MDS-UPDRS items. That rule describes a patient's present state and says nothing about their future. This project builds the probabilistic extension: given what is known about a patient today, the probability that they occupy each subtype at one, two, three and five years.

The work proceeds in three parts. A labelled longitudinal panel of 1,080 sporadic PD patients was built from the PPMI dataset, with the three-class subtype recomputed from raw MDS-UPDRS items and validated at 100.00% against PPMI's own two-class derivation and 99.87% against an archived three-class one. The 3,361 observed transitions in that panel were then characterised. Finally five progression models were fitted and validated on held-out patients.

Three findings shaped everything after them. **Indeterminate is a crossing, not a state** — only 19.2% of patients remain in it. **35.2% of observed subtype changes reverse at the very next visit**, yet the movement is significantly asymmetric ($p = 1.8 \times 10^{-8}$), so the data is a genuine drift seen through a noisy measurement. And **a Markov chain in the observed label is provably misspecified** ($\chi^2 = 156.6$, $p = 7.9 \times 10^{-33}$).

No single model wins. A hidden Markov model with misclassification predicts a patient's next visit best once they have follow-up history, beating the discrete baseline in 10 of 10 splits. The proposal's hierarchical Bayesian growth model supplies the interpretable progression rate — the TD/PIGD ratio declines 0.093 per year (95% CI 0.068–0.116) — but predicts subtype less well and is systematically miscalibrated. For a patient seen for the *first* time, which is the clinical question, both are beaten by a direct model using baseline measures, at 71.9% accuracy.

Every objective set out for this phase of the project has been met. The work is reproducible from a short sequence of commands and is covered by 144 unit tests.

Contents

1 Objective **3**
1.1 Scope . . . 3

2 Constructing a labelled longitudinal dataset **3**
2.1 Sources . . . 3
2.2 Construction . . . 4
2.3 Two data traps . . . 4
2.4 Validation . . . 4
2.5 Result . . . 4

3 What subtype transitions actually look like **4**
3.1 Four findings . . . 5
3.2 Where patients actually end up . . . 7
3.3 Time to PIGD conversion . . . 9
3.4 A bias with a known direction . . . 10

4 Probabilistic models of progression **10**
4.1 A Markov chain in the observed label is misspecified . . . 10
4.2 Model comparison . . . 11
4.3 What the hidden Markov model says . . . 11

5 A hierarchical Bayesian growth model of the ratio **13**
5.1 Two problems in the specification, found by implementing it . . . 13
5.2 Comparison with the hidden Markov model . . . 14

6 The clinical question: a patient seen for the first time **14**
6.1 Example clinical output . . . 15
6.2 Which baseline measures matter . . . 15

7 Exploratory analysis and model validation **15**
7.1 The ratio, examined before any model is fitted . . . 15
7.2 How much of a patient-specific estimate is real . . . 16
7.3 Validation by withholding each patient's future visits . . . 16

8 Which model answers which question **16**

1 Objective

The TD/PIGD classification (Stebbins et al., 2013) computes two subscores from sixteen MDS-UPDRS items and takes their ratio:

$$\text{ratio} = \frac{\text{mean of 11 tremor items}}{\text{mean of 5 gait and postural items}}$$

with ratio $\geq 1.15 \Rightarrow$ TD, ratio $\leq 0.90 \Rightarrow$ PIGD, and the interval between them Indeterminate.

The rule is widely used and clinically validated, but it is a snapshot. Patients move between subtypes as the disease progresses, and the classification carries no statement about how likely that is. The project's objective is to attach probabilities to it without replacing it — to keep the existing thresholds and report, for an individual patient, the chance of remaining where they are or moving.

1.1 Scope

This report covers the first phase of the project: building the dataset, measuring how subtypes change over time, and fitting and validating models that turn those measurements into predictions. Where a result rests on an assumption or carries a known bias, that is stated at the point the result is given rather than collected at the end.

2 Constructing a labelled longitudinal dataset

2.1 Sources

Four files, of which two had to be downloaded separately.

Source	Size	Role
PPMI Curated Data Cut	19,450 × 212	visit skeleton, covariates
MDS-UPDRS Part II (items)	37,777 × 22	3 of the 16 items
MDS-UPDRS Part III (items)	39,310 × 65	13 items, medication state
Archived Apr-2020 cut	3,445 × 159	three-class validation only

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The curated cut alone cannot support the project, which was established before any modelling. It ships `pigd` (the five-item PIGD mean) but no tremor subscore, so the ratio has a denominator and no numerator. Its `td_pigd` is coded *two*-class (TD / non-TD), collapsing Indeterminate away entirely. Both the continuous ratio and the middle class had to be recomputed from raw items.

2.2 Construction

Step	Rows	Removed
Curated cut	19,450	—
Restrict to sporadic PD	5,789	13,661
Drop unscheduled and withdrawal visits	5,774	15
MDS-UPDRS Part III, all rows	39,310	—
Keep OFF-medication or untreated exams	29,356	9,954
Merge with Part II, deduplicate	37,886	11
Panel after joining	**5,774**	
of which carry a subtype label	**4,441**	(76.9%)

Table 1: Every removed row was verified: all 9,954 dropped from Part III have `PDSTATE == 'ON'`, and none are OFF or untreated. The OFF exam is used because levodopa suppresses tremor, so scoring an ON exam would manufacture transitions that are pharmacological rather than pathological.

2.3 Two data traps

Both would have corrupted results silently rather than raising an error.

"Unable to Rate" is stored as the number 101. In all thirteen Part III items, in the same numeric field as a real 0–4 rating. A single 101 among the five PIGD items lifts their mean to roughly 20, driving the ratio to zero and relabelling a tremor-dominant visit as PIGD. `NP3PSTBL` alone carries 3,206 of them among OFF exams. The pipeline excludes values outside the valid 0–4 range, so a sentinel PPMI introduces later cannot enter a subscore either.

The archived validation file uses a different coding. Its `td_pigd_old` defines 2 = PIGD and 3 = Indeterminate, the reverse of the natural ordering. Compared without remapping, agreement reads 65.9% and looks like a broken classifier. Confirmed against the archived data dictionary, remapped, and unit-tested.

2.4 Validation

Reference	Classes	Agreement
Curated cut `td_pigd`	2 (TD / non-TD)	4,431 / 4,431 = **100.00%**
Archived `td_pigd_old`	3 (TD / Indet. / PIGD)	1,595 / 1,597 = **99.87%**

Table 2: The three-class check covers 390 patients and is the only available test of the Indeterminate/PIGD boundary, since the curated cut collapses those two.

2.5 Result

1,080 patients, 5,774 visits, 4,441 labelled, up to 14 years of follow-up. Baseline distribution: TD 71.7%, Indeterminate 10.1%, PIGD 18.1%.

3 What subtype transitions actually look like

3,361 consecutive labelled visit pairs from the 934 patients with at least two labelled visits; median interval 1.00 years.

	→ TD	→ Indeterminate	→ PIGD	Row n
TD	**0.808**	0.080	0.112	2,261
Indeterminate	0.440	**0.192**	0.368	334
PIGD	0.184	0.114	**0.702**	766

Table 3: Pooled transition matrix. All probabilities carry Wilson score intervals in `outputs/transitions/`.

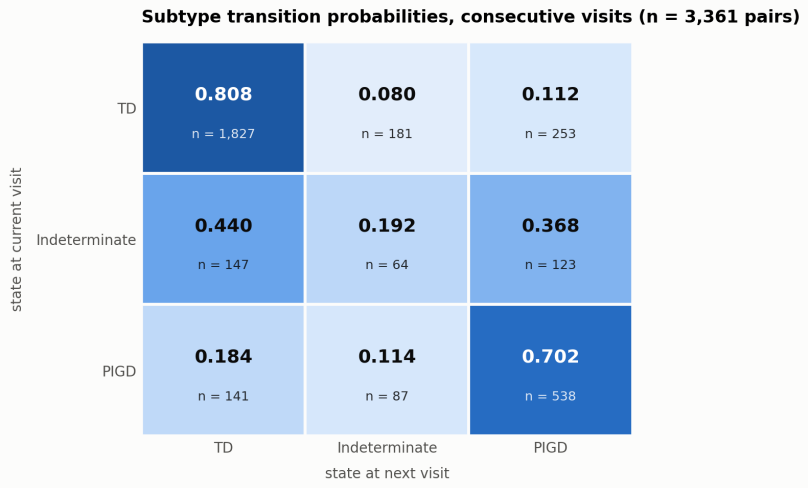


Figure 1: The same matrix. Cell counts are printed because row totals differ by an order of magnitude.

3.1 Four findings

1. Indeterminate is a crossing, not a state. Only 19.2% remain; 44.0% return to TD and 36.8% move to PIGD. Expected persistence 1.24 visits against 5.21 for TD. It behaves exactly as the neutral band of a continuous, noisily measured quantity would — which is the proposal's premise, measured rather than assumed.

2. PIGD is not absorbing. 29.8% of PIGD visits are followed by a non-PIGD assessment, 18.4% straight back to TD. Much of the literature treats TD → PIGD as one-directional; this data does not.

From	To	n	Reverted	Rate
TD	Indeterminate	134	75	**0.560**
PIGD	Indeterminate	54	27	**0.500**
PIGD	TD	112	44	0.393
TD	PIGD	172	55	0.320
Indeterminate	PIGD	93	17	0.183
Indeterminate	TD	108	19	0.176
Overall		**673**	**237**	**0.352**

Table 4: Share of observed state changes that return to the origin at the very next visit. Changes *into* Indeterminate reverse most.

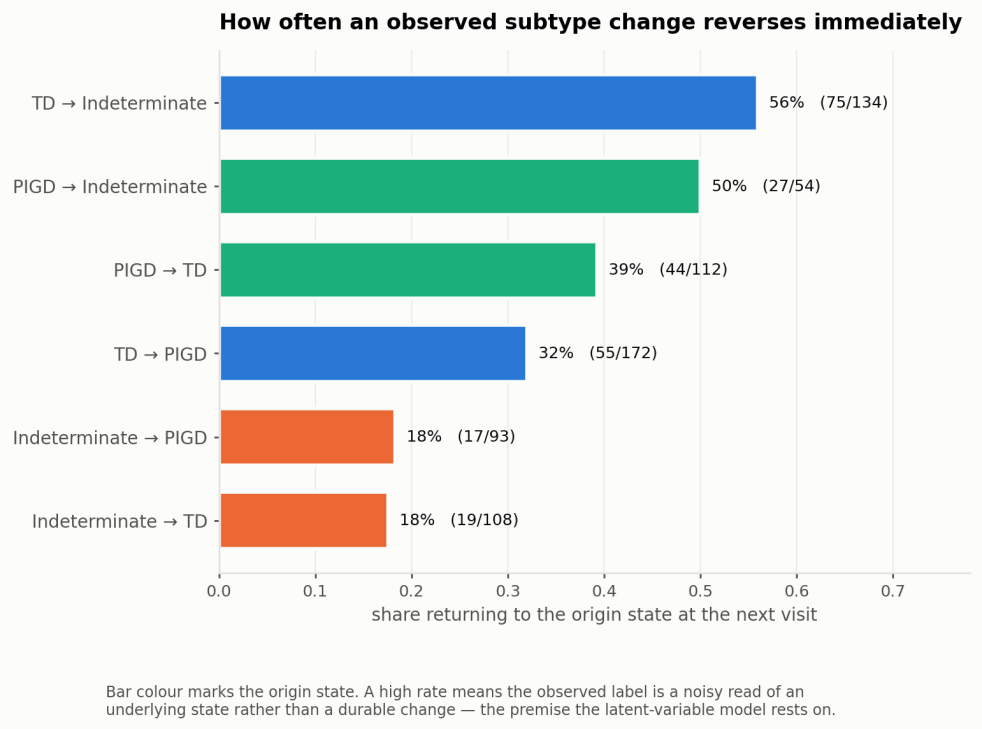


Figure 2: The same rates, ordered. Bar colour marks the state being left. A high rate means the recorded label is a noisy reading rather than a durable change.

3. A third of observed changes reverse immediately.

But the movement is not symmetric, and that distinction carries the project. If the changes were pure noise, forward and reverse counts would match. TD → PIGD occurs 253 times against 141 in reverse ($p = 1.8 \times 10^{-8}$); Indeterminate → PIGD 123 against 87 ($p = 0.016$). Of 932 cross-state moves, 80.5% cancels and 19.5% is net drift toward PIGD.
The correct reading is *a genuine directional drift observed through a noisy measurement* — which is the case *for* a latent-variable model, not against it.

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4. Transition probabilities scale with the gap between visits. TD → PIGD is 0.100 at intervals under 1.5 years and 0.289 beyond four. A single pooled matrix therefore averages one-year and multi-year transitions in whatever proportion attendance happened to produce; it describes the study design as much as the disease.

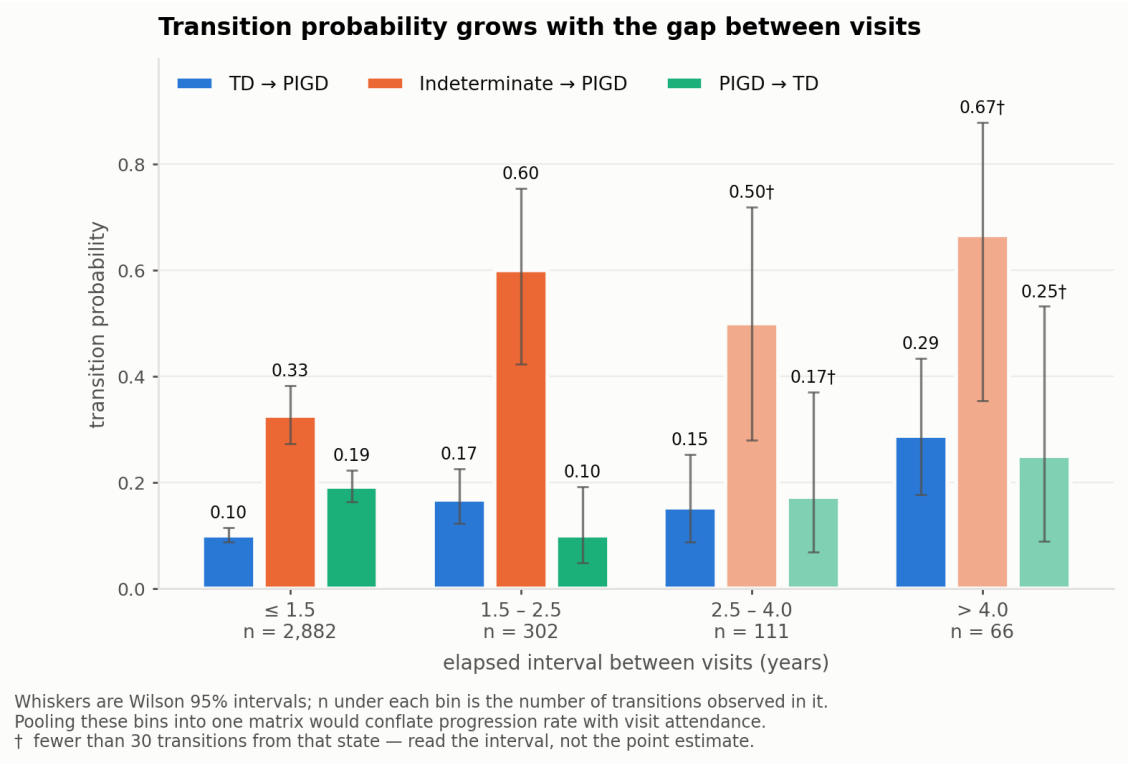


Figure 3: Selected transitions against the elapsed interval, with Wilson 95% intervals. Washed-out bars marked † rest on fewer than 30 observations and should be read as intervals rather than point estimates.

3.2 Where patients actually end up

The matrix above describes single steps. Following the same patients through several visits shows something a one-step matrix cannot.

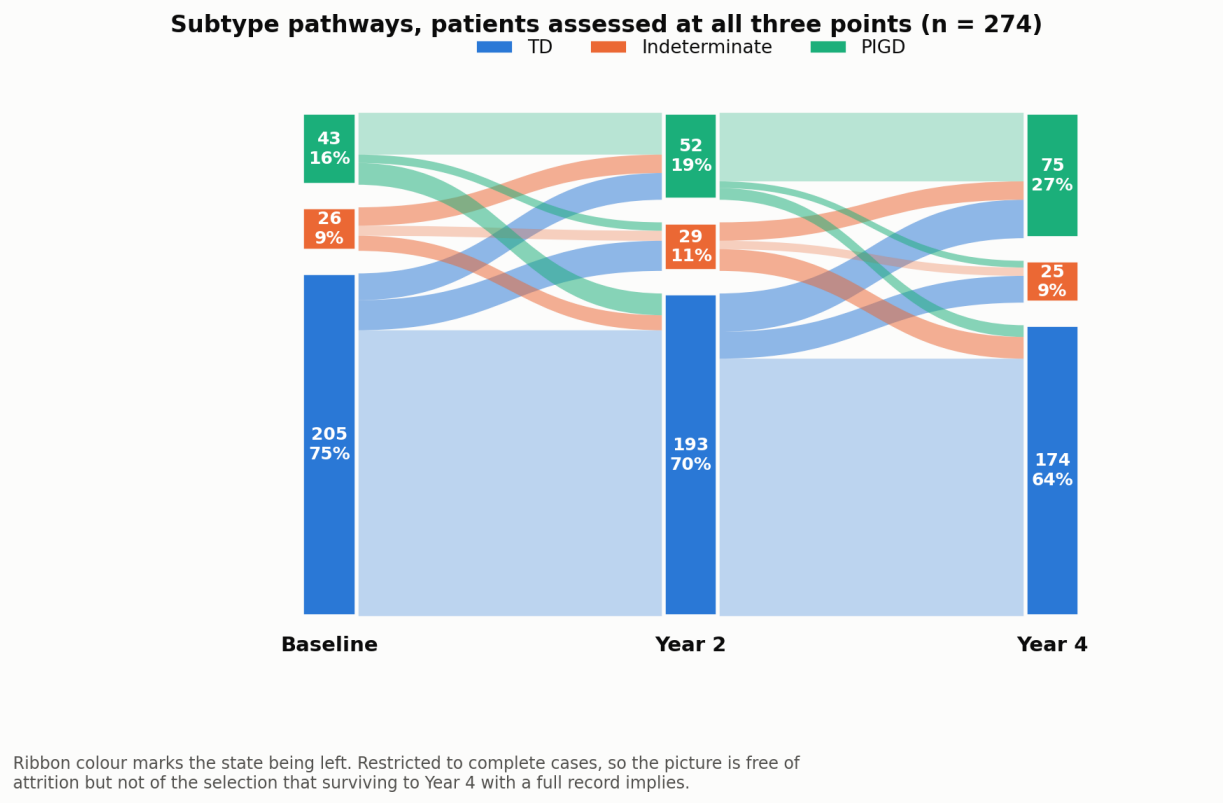


Figure 4: The 274 patients assessed at baseline, year 2 and year 4. Ribbon colour marks the state being left.

Indeterminate looks stable and is not. Its size barely moves — 26 patients at baseline, 29 at year 2, 25 at year 4 — but of the 26 who start there, **only 6 are still there two years later**. Eleven have moved to PIGD and nine back to TD, while 23 different patients have arrived to replace them.

The pool stays the same size with almost entirely different members. A cross-sectional count would show a flat 10% band and suggest a stable clinical category; the flow shows a doorway.

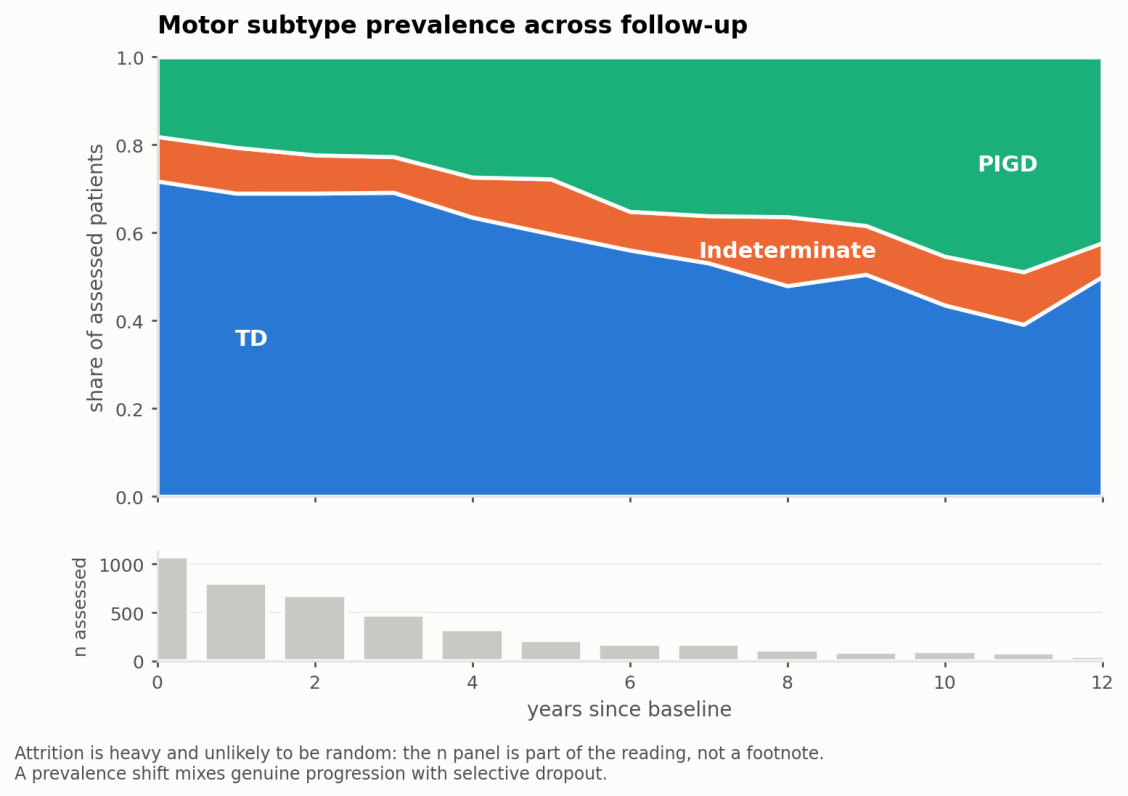


Figure 5: Subtype prevalence across follow-up, with the number of patients actually assessed each year beneath. TD falls from 71.7% to 39.1% and PIGD rises from 18.1% to 48.9% — but by year 8 fewer than 12% of the baseline cohort is still contributing, so this mixes genuine progression with selective dropout. The lower panel is part of the reading, not a footnote.

3.3 Time to PIGD conversion

	Naive (1 visit)	Sustained (2 visits)
At risk (not PIGD at baseline)	885	885
Converted	263 (29.7%)	127 (14.4%)
KM cumulative incidence, year 5	**35.3%**	**15.3%**

Table 5: Requiring a conversion to hold for two consecutive visits eliminates 51.7% of them. The definition of "converted" moves the answer more than most modelling choices downstream of it.

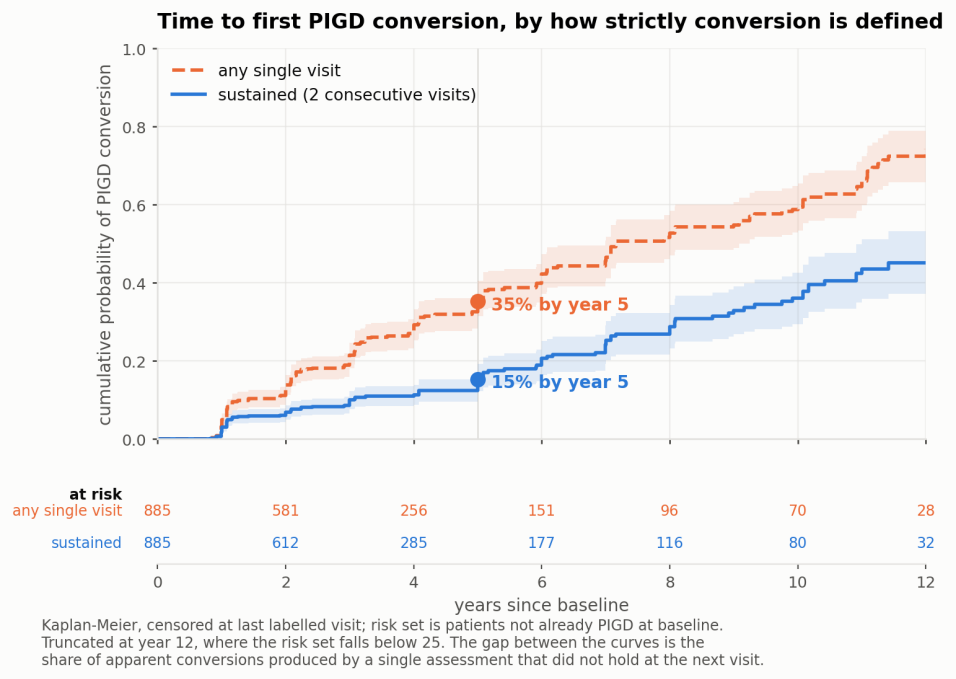


Figure 6: Kaplan–Meier cumulative incidence under both definitions, truncated at year 12 where the risk set falls below 25. The at-risk counts are printed beneath the axis.

3.4 A bias with a known direction

23.1% of cohort visits carry no label. Roughly half are exams that were performed but could not be fully scored — and in **1,243 of those 1,253 exams the missing item is** `NP3PSTBL`, item 3.12, postural stability. The pull test is often unsafe in a patient with poor balance.

Those exams come from materially more advanced patients: median Hoehn & Yahr 3.0 against 2.0 (Mann–Whitney $p = 1.4 \times 10^{-17}$). Postural stability is one of the five *PIGD* items, so the visits lost are disproportionately those of patients most likely to be classified PIGD.

Every progression estimate in this report is therefore a **lower bound**. Relaxing the all-items rule is not a remedy: allowing one missing item recovers 44 visits and changes no existing label, because the incomplete exams are missing many items rather than one.

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4 Probabilistic models of progression

Five models were fitted and validated on held-out patients. Splitting is always at the level of the *patient*, stratified by baseline subtype, so no individual appears on both sides.

4.1 A Markov chain in the observed label is misspecified

A Markov chain assumes the current state summarises the past. That is directly testable on the 2,427 labelled triples.

Currently	Triples	χ^2 (dof 4)	p
TD	1,583	156.6	7.9×10^{-33}
Indeterminate	235	15.3	4.2×10^{-3}
PIGD	609	61.2	1.7×10^{-12}

For visits currently labelled TD, arriving from PIGD raises the probability of being PIGD next visit from **0.086 to 0.393**. Where a patient came from matters enormously, even conditioning on where they are. That is the signature of a latent state read through noise: if the true phenotype is PIGD but one exam scores TD, the next exam is likely to score PIGD again.

4.2 Model comparison

Model	Log-lik / transition	Accuracy	Brier
HMM (latent state + misclassification)	**−0.6384**	**0.770**	**0.3458**
Discrete Markov chain (per visit)	−0.6653	0.768	0.3625
Continuous-time Markov chain	−0.6752	0.766	0.3670
Marginal (null)	−0.8633	0.625	0.5185

Table 6: 276 held-out patients, 1,002 transitions.

The hidden Markov model beats the discrete baseline in **10 of 10** independent splits, by $+0.0284 \pm 0.0094$ log-likelihood per transition (paired t-test $p = 5.4 \times 10^{-6}$, Wilcoxon $p = 0.002$).
The continuous-time chain — the expected improvement, given that transition probabilities scale with the interval — **does not help**. It loses at every interval and by most at long ones. Imposing $P(t) = e^{Qt}$ enforces a consistency across intervals that a noisily observed process does not satisfy, so a correct-looking constraint on a misspecified model makes it worse.

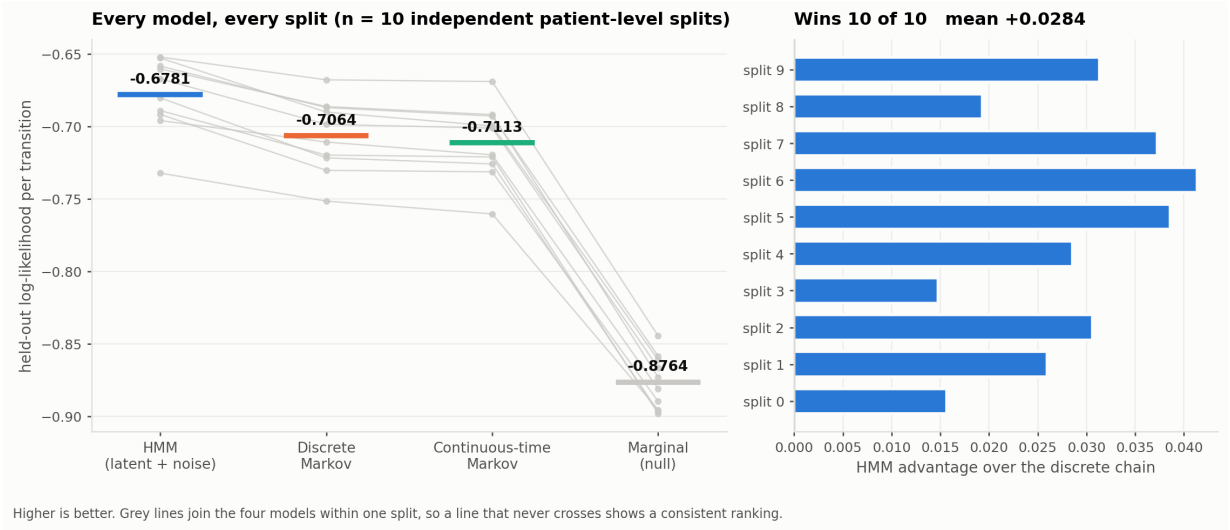


Figure 7: Every model on every split. Grey lines join the four models within one split, so a line that never crosses shows a consistent ranking.

4.3 What the hidden Markov model says

From	→ TD	→ Indeterminate	→ PIGD
TD	−0.0994	0.0937	**0.0057**
Indeterminate	0.1156	−0.4713	0.3557
PIGD	**0.0000**	0.0605	−0.0605

Table 7: Latent generator Q, rates per year.

Progression is sequential. The direct TD → PIGD rate is 0.0057 per year against 0.0937 for TD → Indeterminate — **16 times smaller** — and direct PIGD → TD is estimated at exactly zero. The model was never told the states are ordered; it recovered that. This is what a continuous underlying ratio implies: to cross from above 1.15 to below 0.90 a patient must pass through the middle band.

The label is unreliable in a specific place. The model estimates how often an examination records each true phenotype correctly:

True state	recorded TD	recorded Indet.	recorded PIGD
TD	**0.941**	0.038	0.021
Indeterminate	0.250	**0.500**	0.250
PIGD	0.106	0.052	**0.842**

Table 8: Estimated misclassification. The Indeterminate row rests on an imposed floor of 0.50, so it reads as "at most half of Indeterminate labels are correct". This independently corroborates the finding in Section 3 that changes into Indeterminate reverse most often (50–56%), by an entirely different route.

The phenotype is more stable than its label. Separating measurement error from genuine change recovers a much slower underlying process:

State	Observed label	Latent phenotype
TD	3.41	**10.06**
Indeterminate	0.44	**2.12**
PIGD	2.28	**16.52**

Table 9: Mean years before leaving a state. Treating the label as the state makes every phenotype look transient.

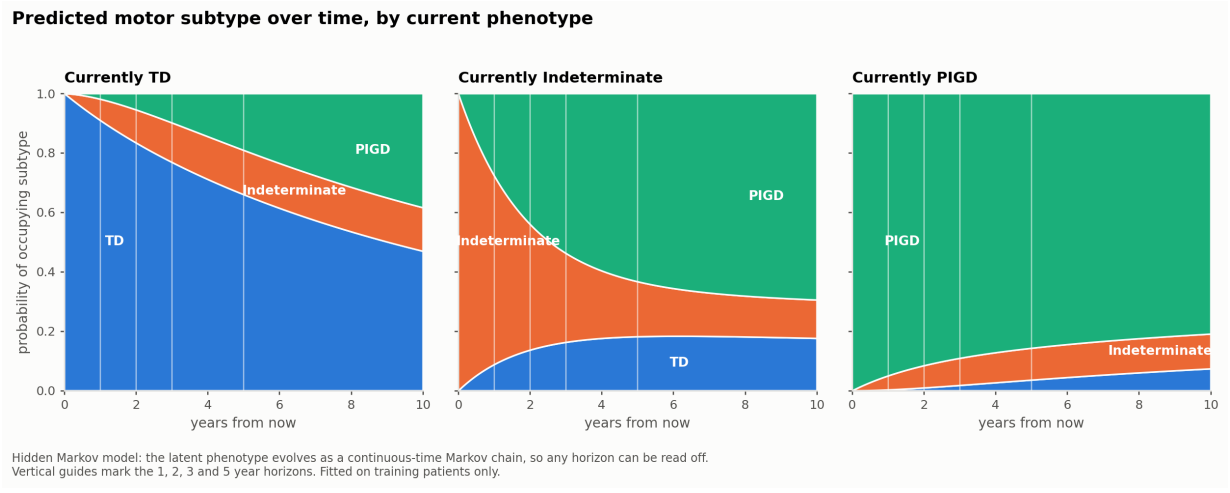


Figure 8: Predicted subtype over time by current phenotype. Because the latent process runs in continuous time, any horizon can be read off.

5 A hierarchical Bayesian growth model of the ratio

Implemented as specified (proposal pp. 13–15):

$$R_{it}^{obs} = R_{it} + \epsilon_{it}^{obs}, \qquad R_{it} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\, t + \epsilon_{it}^{proc}$$
$$b_{0i} \sim \mathcal{N}(0, \tau_0^2), \qquad b_{1i} \sim \mathcal{N}(0, \tau_1^2)$$

Inference by Gibbs sampling with conjugate updates throughout; roughly one second over 573 patients, validated by recovering known parameters from simulated data.

Parameter	Mean	2.5%	97.5%
β_0 population baseline ratio	2.2920	2.1577	2.4272
β_1 population progression, /yr	−0.0927	−0.1159	−0.0682
σ total residual SD	1.1380	1.0993	1.1778
τ_0 SD of patient baselines	1.3408	1.2466	1.4421
τ_1 SD of patient slopes, /yr	0.1033	0.0780	0.1315

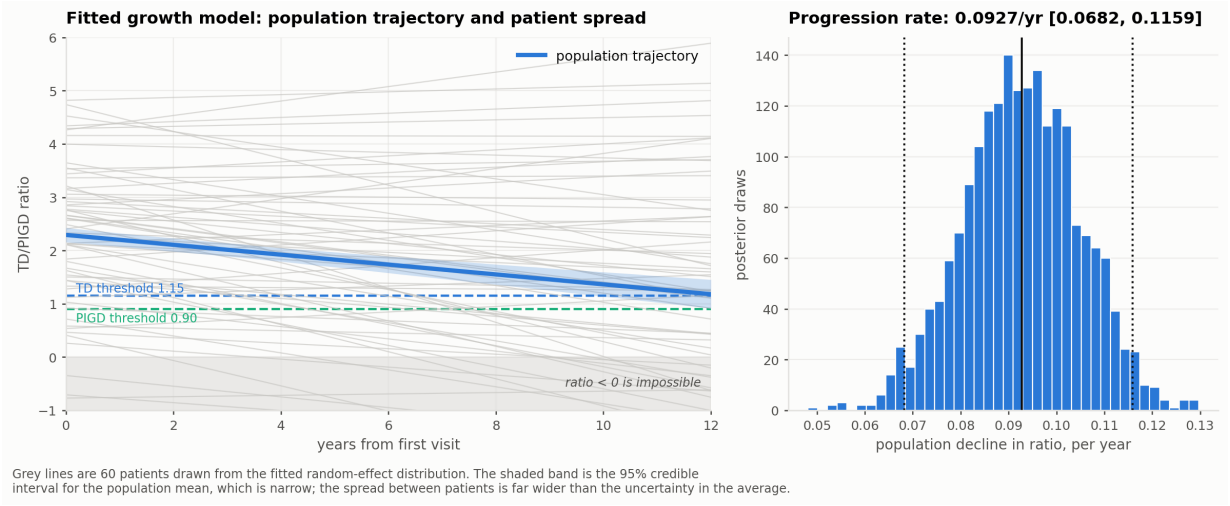


Figure 9: Population trajectory with 60 patients drawn from the fitted random-effect distribution, and the posterior for the progression rate.

The ratio declines 0.0927 per year, $P(\beta_1 < 0) = 1.000$, robust across five very different prior specifications (τ_1 moves only within 0.094–0.116). From the mean baseline of 2.292 the population average reaches the TD threshold after a median 12.3 years.

5.1 Two problems in the specification, found by implementing it

The two variances are not separately identifiable. Substituting the growth model into the observation model,

$$R_{it}^{obs} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\, t + \underbrace{(\epsilon_{it}^{proc} + \epsilon_{it}^{obs})}_{\text{variance } \sigma_{proc}^2 + \sigma_{obs}^2}$$

Both terms are additive on the same observation, so the likelihood depends only on their sum — verified numerically. Only $\sigma^2 = \sigma_{proc}^2 + \sigma_{obs}^2$ is estimated, and this is stated rather than a split reported that the data cannot support. It matters because $P(TD) = P(R^{true} > 1.15)$ refers to the latent ratio, whose spread excludes observation noise.

A Normal likelihood cannot represent the ratio , which is non-negative, right-skewed (skew 1.27) and has an atom at exactly zero (7.0% of visits). The model places **12.2% of its five-year predictive mass below zero**.

5.2 Comparison with the hidden Markov model

Model	Log-lik / transition	Accuracy	Brier
HMM (latent states)	**−0.6559**	**0.761**	**0.3573**
Hierarchical Bayesian growth	−0.6814	0.737	0.3786

Table 10: 825 shared held-out transitions. Over six independent splits the HMM is ahead in 6 of 6, mean +0.0579 (sd 0.0188), paired t-test $p = 6.6 \times 10^{-4}$.

The reason is measurable, not a matter of model capacity. The growth model's residual SD is $\sigma = 1.138$; the Indeterminate band is 0.25 wide. **The noise is 4.6 times the width of the band the model must predict membership of.** Even a perfectly estimated trajectory places the next observation almost anywhere.

The two models are complementary rather than competing. Only the growth model supplies an interpretable progression rate with a credible interval, patient-specific slopes, and a route to the σ_i stability phenotype in the proposal's second half.

6 The clinical question: a patient seen for the first time

Every model above earns its accuracy by filtering a patient's *history* forward. A new patient has none. Measured directly, over five splits, predicting subtype at horizon h from a first visit only:

Model (baseline information only)	Held-out log-likelihood
Baseline subtype + ratio + covariates	**−0.7550**
+ covariates	−0.7625
+ baseline ratio	−0.7735
Baseline subtype + horizon	−0.7813
Hidden Markov model (nothing to filter)	−0.7869
Carry forward ("assume no change")	−1.4114

On the first-visit question the hidden Markov model is the worst of the five, not because it is a poor model but because it has nothing to work with. A direct model of P(subtype at h | first visit) beats it by +0.032, at **71.9% accuracy**, and is well calibrated across the full probability range.

The HMM is not superseded — its advantage over a lookup table grows with history: +0.005 at one visit, +0.021 at two, +0.047 at six or more. Which model to use is decided by how much follow-up exists.

6.1 Example clinical output

Horizon	P(TD)	P(Indeterminate)	P(PIGD)
now	1.000	0.000	0.000
1 year	0.821	0.094	0.086
2 years	0.785	0.105	0.110
3 years	0.749	0.113	0.138
5 years	0.671	0.122	0.207

Table 11: A new TD patient with an otherwise average baseline profile. This is the output format the proposal specifies.

6.2 Which baseline measures matter

Measure	Cost of dropping	Measure	Cost of dropping
MDS-UPDRS Part II	+0.0153	MoCA	−0.0005
DaTscan putamen	+0.0050	UPSIT	−0.0008
Age	+0.0042	Disease duration	−0.0010
Hoehn & Yahr	+0.0021	SCOPA-AUT	−0.0014
Sex	+0.0002	RBDSQ	−0.0016

Table 12: Held-out cost of removing each covariate, one at a time.

The four non-motor markers the proposal's second half is built on — RBDSQ, SCOPA-AUT, UPSIT and MoCA — all sit at or below zero. The covariate gain comes from the motor and imaging measures.
This is *not* a refutation of the proposal's hypothesis, which concerns trajectory *stability* (σ_i) rather than the trajectory mean, and leave-one-out understates a predictor that is redundant given another. But it is worth establishing before that half of the project is built on them.

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7 Exploratory analysis and model validation

7.1 The ratio, examined before any model is fitted

- **Within-patient SD 1.22** against an Indeterminate band 0.25 wide. Band membership was always going to be noise-dominated, and this is visible before a model is fitted.
- **ICC 0.536** — 54% of ratio variance is between patients.
- Raw per-patient OLS slopes have SD **0.531** against the hierarchical τ_1 of 0.10, a factor of five. Shrinkage does real work, and a prior on τ_1 must not be centred on the naive spread.

One result argues against the modelling choice. Lag-1 residual autocorrelation is −0.219. Read against zero this looks like mean reversion, but it cannot be read against zero: residuals from a line fitted to a handful of points are negatively correlated by construction. Simulating iid noise on the same visit pattern gives a null of **−0.332** [−0.365, −0.294].

The observed value is *more positive* than the null and outside it, so deviations persist between visits — the signature of accumulating process noise. That favours the **state-space** form over the growth form, and the state-space form is also where σ_{obs} and σ_{proc} become separately identifiable. Two independent reasons to prefer it.

7.2 How much of a patient-specific estimate is real

Visits	2	3	5	8	10+
Patients	191	233	71	40	50
Median span (yr)	1.16	2.09	4.17	8.42	11.67
Slope reliability	**0.008**	**0.024**	0.112	0.384	0.569
Baseline reliability	0.737	0.804	0.854	0.857	0.877

Table 13: Reliability is $1 - \mathrm{Var}_{post}(b)/\tau^2$: zero means the estimate is the population mean with the patient contributing nothing.

Median slope reliability is 0.050 and no patient exceeds 0.8. For the typical patient, 95% of their "individual progression rate" is the cohort average. The *baseline level*, by contrast, is well identified for essentially everyone (median 0.805).
So the defensible clinical claim is that a prediction is individualised **by position, not by rate**: we know where this patient is, and assume they move at the cohort's pace. Held-out, that individualisation is worth +0.20 log-likelihood, of which 95% comes from the baseline level.

7.3 Validation by withholding each patient's future visits

Withholding each patient's future visits, which is the design the proposal specifies:

Train on first	Future visits	Log-lik	Accuracy	95% coverage
1 visit	2,707	−1.0341	0.576	0.974
2 visits	1,886	**−0.8917**	0.647	**0.929**
3 visits	1,256	−0.9398	0.666	0.905

Interval coverage is close to nominal, so the model's stated uncertainty is honest. Its *calibration* is not: the growth model under-predicts PIGD in every bin (0.018 against 0.078 observed, rising to 0.535 against 0.729), mean absolute gap 0.105. The bias is systematic, one-directional, and unique to this model among the three.

8 Which model answers which question

Question	First-visit	HMM	Growth
New patient, no follow-up	**best**	worst	middle
Patient with several visits	—	**best**	weaker
Progression rate on the clinical scale	no	no	**yes**
Patient-specific progression rate	no	no	**yes**
Quantifies label misclassification	no	**yes**	no
Uses visits with an undefined ratio (16.3%)	**yes**	**yes**	no
Well calibrated	**yes**	**yes**	no
Route to the σ_i phenotype	no	no	**yes**

No single model dominates, and the project should keep all three. The clinical instrument is the first-visit model; the HMM takes over as follow-up accrues; the growth model is the scientific description of progression and the foundation for the proposal's second half.

Reference. Stebbins GT, Goetz CG, Burn DJ, Jankovic J, Khoo TK, Tilley BC. How to identify tremor dominant and postural instability/gait difficulty groups with the Movement Disorder Society Unified Parkinson's Disease Rating Scale. *Movement Disorders* 2013;28(5):668–70.